Fine-Tuning an LLM using LoRA/QLoRA

1. Install the required libraries: transformers, datasets, peft, trl, accelerate, and bitsandbytes.
2. Check whether a GPU is available and identify its type and memory.
   Record: What GPU is available? Why is a GPU useful for LLM fine-tuning?
3. Choose a small, instruction-tuned causal language model that is compatible with your available GPU and the current Hugging Face tools.
   Possible models to investigate include:
   Qwen/Qwen2.5-0.5B-Instruct
   HuggingFaceTB/SmolLM2-360M-Instruct
4. Create a small dataset of at least 30 question-and-answer pairs about specific domain (Example: A computer shop). You may create your own dataset or use a suitable public dataset from Hugging Face Datasets.
5. Use the PEFT library to configure LoRA. If GPU memory is limited and the selected model supports it, use QLoRA with 4-bit quantity.
6. Use a supported supervised fine-tuning workflow, such as Hugging Face TRL's SFTTrainer.
7. Test both the original model and the fine-tuned model using the same questions. Include questions that were not present in the training dataset.
8. Test both the original model and the fine-tuned model using the same questions. Include questions that were not present in the training dataset.